

List of cybersecurity information technologies

This is a **list of cybersecurity information technologies**. [Cybersecurity](#) concerns all technologies that store, manipulate, or move computer [data](#), such as [computers](#), [data networks](#), and all devices connected to or included in said networks, such as [routers](#) and [switches](#). All [information technology](#) devices and facilities need to be secured against [intrusion](#), unauthorized use, and [vandalism](#). Users of information technology are to be protected from theft of assets, [extortion](#), [identity theft](#), loss of privacy, damage to equipment, [business process](#) compromise, and general disruption. The public should be protected against acts of [cyberterrorism](#), such as compromise or [denial of service](#).

Cybersecurity is a major endeavor in the [IT industry](#). There are a number of [professional certifications](#) given for cybersecurity training and [expertise](#).^[1] Billions of dollars are spent annually on cybersecurity, but no computer or network is immune from attacks or can be considered completely secure.

This article attempts to list important Wikipedia articles about cybersecurity.

General

Introductory articles about cybersecurity subjects:

- [Security](#)
- [Computer security](#)^[2]
- [Internet security](#)^[3]
- [Network security](#)^[4]
- [Information security](#), [Data security](#)^[5]
- [List of computer security certifications](#)

Cryptography

The art of secret writing or code. A "plaintext" message is converted by the sender to "ciphertext" by means of a mathematical [algorithm](#) that uses a secret key. The receiver of the message then reverses the process and converts the ciphertext back to the original plaintext.^[6]

Cryptography subject matter

- [Ciphertext](#)
- [Encryption](#)
- [Decryption](#)
- [History of cryptography](#)
- [Alan Turing](#)
- [Cipher](#)
- [Plaintext](#)

- [Cryptanalysis](#)
- [Cryptographic primitive](#)
- [Cryptographic Service Provider](#)
- [HMAC](#)
- [HMAC-based One-time Password algorithm](#)
- [Cryptographic hash function](#)
- [Hash collision](#)
- [Hash-based cryptography](#)
- [Cryptographic nonce](#)
- [Salt \(cryptography\)](#)
- [Cryptographic strength](#)
- [Block cipher](#)
- [Block cipher mode of operation](#)
- [Stream cipher](#)
- [Key \(cryptography\)](#)
- [Key size](#)
- [Cryptographic key types](#)
- [Symmetric-key cryptography](#)
- [Public-key cryptography](#) (sometimes called *Asymmetric-key cryptography*)
- [Public-Key Cryptography \(conference\)](#)
- [Digital signature](#)
- [Non-repudiation](#)
- [Public key certificate](#)
- [Certificate authority](#)
- [Public key fingerprint](#)
- [Secret sharing](#)
- [Internet key exchange](#)
- [Strong cryptography](#)
- [Brute-force attack](#)
- [Dictionary attack](#)
- [Padding oracle attack](#)
- [Pass the hash](#)
- [Substitution cipher](#)
- [One-time pad](#)
- [Beale ciphers](#)
- [The Codebreakers^{\[7\]}](#)
- [Data Encryption Standard](#)
- [Advanced Encryption Standard](#)
- [International Data Encryption Algorithm](#)
- [List of hash functions](#)
- [Comparison of cryptographic hash functions](#)
- [SHA-1](#)
- [SHA-2](#)
- [SHA-3](#)
- [SHA-3 competition](#)
- [RSA \(cryptosystem\)](#)
- [X.509](#)
- [Pretty Good Privacy](#)
- [Diffie-Hellman key exchange](#)
- [Blowfish \(cipher\)](#)

Cipher technologies

- [Enigma machine](#)
- [Caesar Cipher](#)
- [Vigenère cipher](#)

Steganography

[Steganography](#) is the process of hiding data within other data, most commonly by hiding data inside images.^[8]

- [BPCS-Steganography](#)
- [Steganography tools](#)
- [Steganalysis](#)
- [OpenPuff](#)
- [Kristie Macrakis^{\[9\]}](#)

Authentication and access

The process by which a potential client is granted authorized use of an IT facility by proving its identity.^[10]

- Authentication
- Login
- Password
- Passphrase
- Password strength
- One-time password
- Multi-factor authentication
- Identity management
- Identity management theory
- Identity management system
- Encrypting PIN Pad
- Shared secret
- Authorization
- Access control
- Principle of least privilege
- Cryptographic protocol
- Authentication protocol
- Public key infrastructure
- RADIUS
- Kerberos (protocol)
- OpenID
- OAuth
- Active Directory Federation Services
- Security Assertion Markup Language
- SAML-based products and services

Public Key Infrastructure (PKI)

A framework for managing digital certificates and encryption keys.

- Public key infrastructure
- X.509
- Root certificate
- Public key certificate
- Certificate authority
- Digital signature
- Certificate policy
- Certificate Practice Statement
- Certificate revocation list
- Online Certificate Status Protocol

Tools

Computerized utilities designed to study and analyze the security of IT facilities and/or break into them on an unauthorized and potentially criminal basis.^[11]

- List of security assessment tools
- Kali
- Security Administrator Tool for Analyzing Networks
- Nessus (software)
- Vulnerability scanner
- Nessus Attack Scripting Language

- [OpenVAS](#)
- [Yasca](#)
- [Metasploit project](#)
- [John the Ripper](#)
- [Smeg Virus Construction Kit](#)
- [Virus Creation Laboratory](#)
- [Exploit kit](#)

Threats

Modes of potential attacks on IT facilities.^[12]

- [Cyberattack](#)
- [STRIDE \(security\)](#)
- [Vulnerability \(computing\)](#)
- [Common Vulnerabilities and Exposures](#)
- [Privilege escalation](#)
- [Social engineering \(security\)](#)
- [Malware](#)
- [Spyware](#)
- [Backdoor \(computing\)](#)
- [Computer virus](#)
- [Computer worm](#)
- [Macro virus](#)
- [Keystroke logging](#)
- [Trojan horse](#)
- [Hardware Trojan](#)
- [Eavesdropping](#)
- [Zombie](#)
- [Botnets](#)
- [Advanced persistent threat](#)
- [Man-in-the-middle attack](#)
- [Man-on-the-side attack](#)
- [Meet-in-the-middle attack](#)
- [Length extension attack](#)
- [Replay attack](#)
- [Pre-play attack](#)
- [Dictionary attack](#)
- [Biclique attack](#)
- [Denial-of-service attack](#)
- [Resource exhaustion attack](#)
- [Brute-force attack](#)
- [Watermarking attack](#)
- [Mangled packet](#)
- [Reverse connection](#)
- [Polymorphic code](#)
- [Password cracking](#)
- [Spoofing attack](#)
- [POODLE](#)

Exploits

Security exploits affecting computers.^[13]

- [Exploit \(computer security\)](#)
- [Timeline of computer viruses and worms](#)
- [Comparison of computer viruses](#)
- [Malware analysis](#)
- [XML denial-of-service attack](#)
- [Distributed denial-of-service attacks on root nameservers](#)
- [Linux malware](#)
- [Zero-day \(computing\)](#)
- [Virus hoax](#)
- [Pegasus](#)
- [Rogue security software](#)
- [MS Antivirus \(malware\)](#)